Self-Introduction

Zifei Bai

March 19th, 2026

UNIVERSITY OF MICHIGAN

Education and Background

- Computer Science and Data Science @ UMich, 2023-2026
- Previous research experiences on transformer-based language models and their generalization problems
- Peer Advisor @ Department of Statistics, University of Michigan

Meet your Peer Advisors!



Zifei Bai

Zifei Bai

Hi, my name is Zifei, and I'm a senior majoring in Computer Science and Data Science. On campus, I have several research experiences, including URPS and under the EECS department. Outside of academics, I love travelling and hiking. If you are particularly interested in finding research opportunities, I am happy to discuss it. I'm excited to meet and talk with all of the students!

Graph from:
https://lsa.umich.edu/stats/undergraduate-students/advising/undergrad-peer-advising.html

Research Experience

SLIDING CRITICAL BAND IN ROPE-BASED LENGTH EXTRAPOLATION

DeLTa @ ICLR 2026
link

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ABSTRACT

Context extension in RoPE-based Large Language Models has become a primary focus in the development of RoPE-based models. In this paper, we introduce Sliding Critical Band, a framework demonstrating that the dimensions requiring interpolation dynamically migrate across the spectrum under different extrapolation ratios. Building on this, we proposed Spectrum Bandwidth Exhaustion, which provides an explanation of why larger RoPE bases can enhance models' extrapolation ability. Together, these two concepts offer a more comprehensive understanding of the principles underlying context extrapolation in RoPE-based models. Evaluations on synthetic tasks and the C4 dataset validate the universality of the Sliding Critical Band across diverse scenarios.

Outline

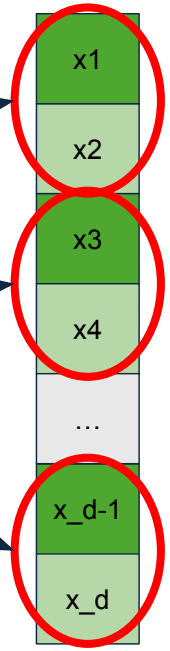
- Related Works and Motivation
- Formalizing *Critical Band* Concept
- Introducing *Spectrum Bandwidth Exhaustion*
- Further Validation
- Summary

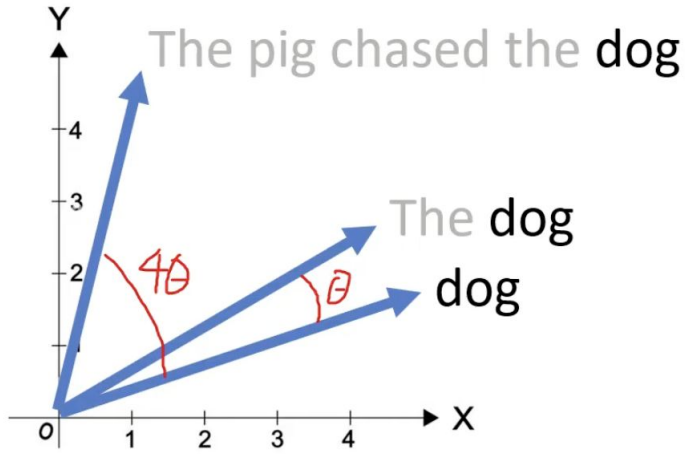
Related Works and Motivation

Rotary Positional Encoding (Su et al., 2023)

$$f_{\{q,k\}}(\boldsymbol{x}_m, m) = \boldsymbol{R}^d_{\Theta,m} \boldsymbol{W}_{\{q,k\}} \boldsymbol{x}_m$$

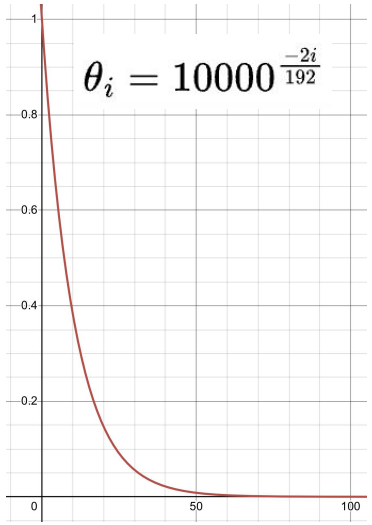
$$\boldsymbol{R}^d_{\Theta,m} = \begin{pmatrix} \cos m\theta_1 & -\sin m\theta_1 & 0 & 0 & \cdots & 0 & 0 \\ \sin m\theta_1 & \cos m\theta_1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cos m\theta_2 & -\sin m\theta_2 & \cdots & 0 & 0 \\ 0 & 0 & \sin m\theta_2 & \cos m\theta_2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & \cos m\theta_{d/2} & -\sin m\theta_{d/2} \\ 0 & 0 & 0 & 0 & \cdots & \sin m\theta_{d/2} & \cos m\theta_{d/2} \end{pmatrix}$$





$$\theta_i = \theta^{-2i/d}$$

Larger i, lower frequency



Graph from: https://www.youtube.com/watch?v=o29P0Kpobz0

Related Works and Motivation

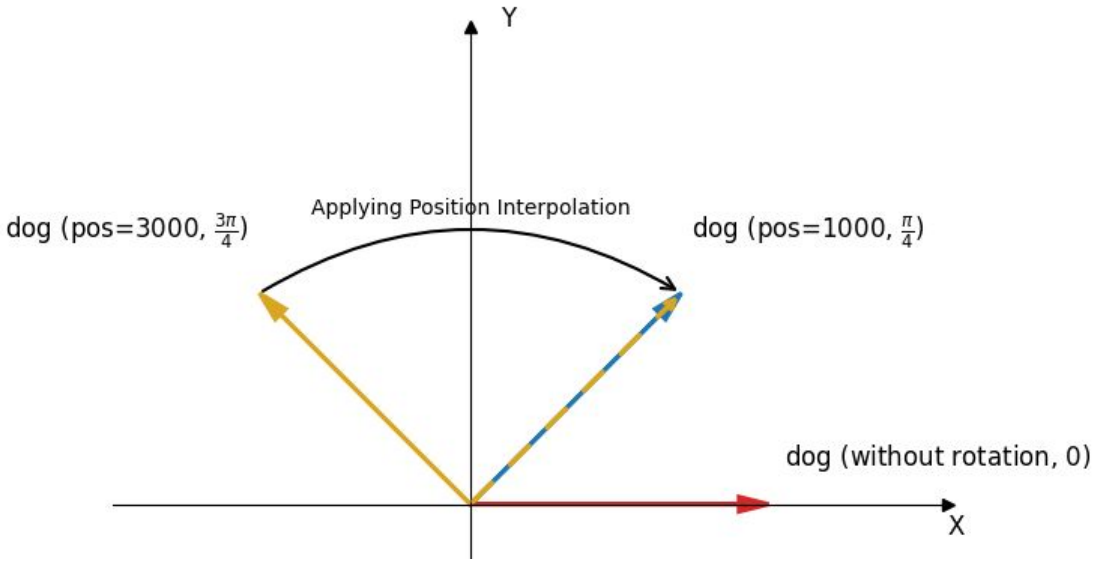
Common Belief for Failure of RoPE-based Context Extrapolation (Peng et al., 2023, Liu et al., 2024)

Dimensions that **fail to complete a full rotation cycle** during training will enter OOD angle during long-context scenario. (i.e. dimensions with higher index and lower frequency)

$$\frac{2\pi}{\theta^{-2d_{\text{extra}}/d_{\max}}} = 2\pi \cdot \theta^{\frac{2d_{\text{extra}}}{d_{\max}}} > L_{\text{train}} \quad \Longrightarrow \quad d_{\text{extra}} = \left\lceil \frac{d_{\max}}{2} \log_{\theta}\left(\frac{L_{\text{train}}}{2\pi}\right) \right\rceil$$

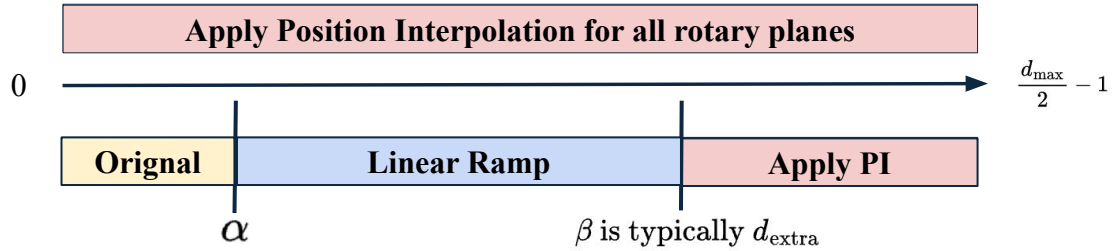
Related Works and Motivation

RoPE-based Context Extension (Chen et al., 2023, Peng et al., 2023)



For simplicity, we denote the RoPE transformation for i-th subvector $\mathbf{x} \in \mathbb{R}^2$ as a function of $\mathfrak{R}_i(\mathbf{x}, m)$

PI (Chen et al., 2023): $\mathfrak{R}_i\left(\mathbf{x}, \frac{m}{s}\right)$ for $i \in \left[0, \frac{d}{2} - 1\right]$ $\quad s = \frac{L'}{L_{\text{train}}}$



YaRN (Peng et al., 2023):

$$\mathfrak{R}_i^*(\mathbf{x}, m) = \begin{cases} \mathfrak{R}_i(\mathbf{x}, m) & \text{if } i \in [0, \alpha] \\ \mathfrak{R}_i\left(\mathbf{x}, \frac{m}{s} \cdot \frac{i-\alpha}{\beta-\alpha}\right) & \text{if } i \in (\alpha, \beta) \\ \mathfrak{R}_i(\mathbf{x}, \frac{m}{s}) & \text{if } i \in [\beta, \frac{d}{2} - 1] \end{cases}, \quad s = \frac{L'}{L_{\text{train}}}; \quad \alpha, \beta \text{ case by case}$$

performs well on s=4, 8, 16 by finetuning

Related Works and Motivation

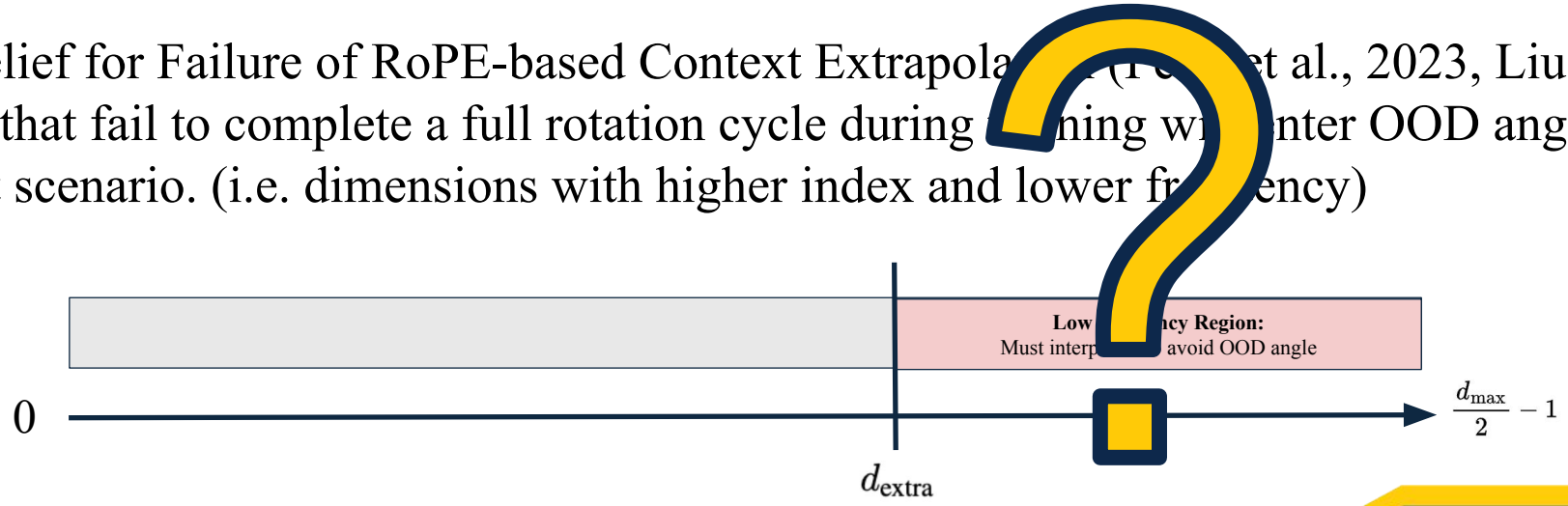
🌟Discrepancy:

Interpolating all the RoPE dimensions that $d \geq d_{\text{extra}}$ is not enough

especially under mild extrapolation ratio s ($s = 1.2, 1.5$)

Following the common belief, we interpolate all the rotary dimensions that did not complete a full cycle, compress them into in-distribution angle, but **still performs bad during long-context**.

Common Belief for Failure of RoPE-based Context Extrapola (Re et al., 2023, Liu et al., 2024): Dimensions that fail to complete a full rotation cycle during ining wi nter OOD angle during long-context scenario. (i.e. dimensions with higher index and lower fr ency)



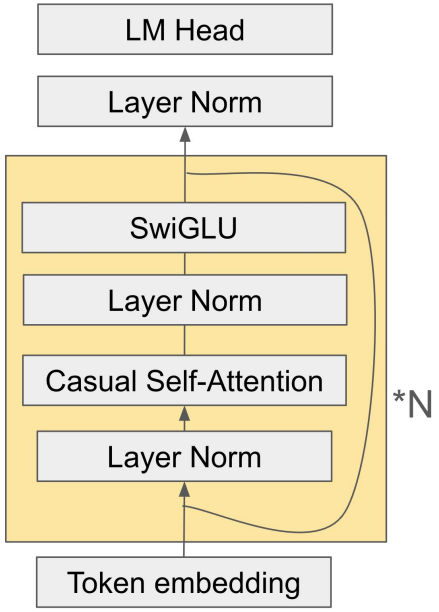
Preliminaries

Task

Table 1: Examples of string-copying task

Input (Q: prompt, A: label)		Task difficulty
Q: 12345=	A: 12345	Length of string
Q: 123456789=	A: 123456789	

Model



4-layer, 2-head, 384 embedding dimensions

7.1M parameters, RoPE-base=10,000

Preliminaries

Data

$$l \sim U[1, l_0]$$

Generate random sequence with length of l

$$\mathcal{D}_{\text{train}} = \{(x_i, y_i)\}_{i=1}^{N}, \quad \text{where Length}(x_i) \leq l_0$$

Metric

Accuracy, Perplexity

Extrapolation Ratio (s)

$$s = \frac{L'}{L_{\text{train}}}$$

Preliminaries

Interpolation Method

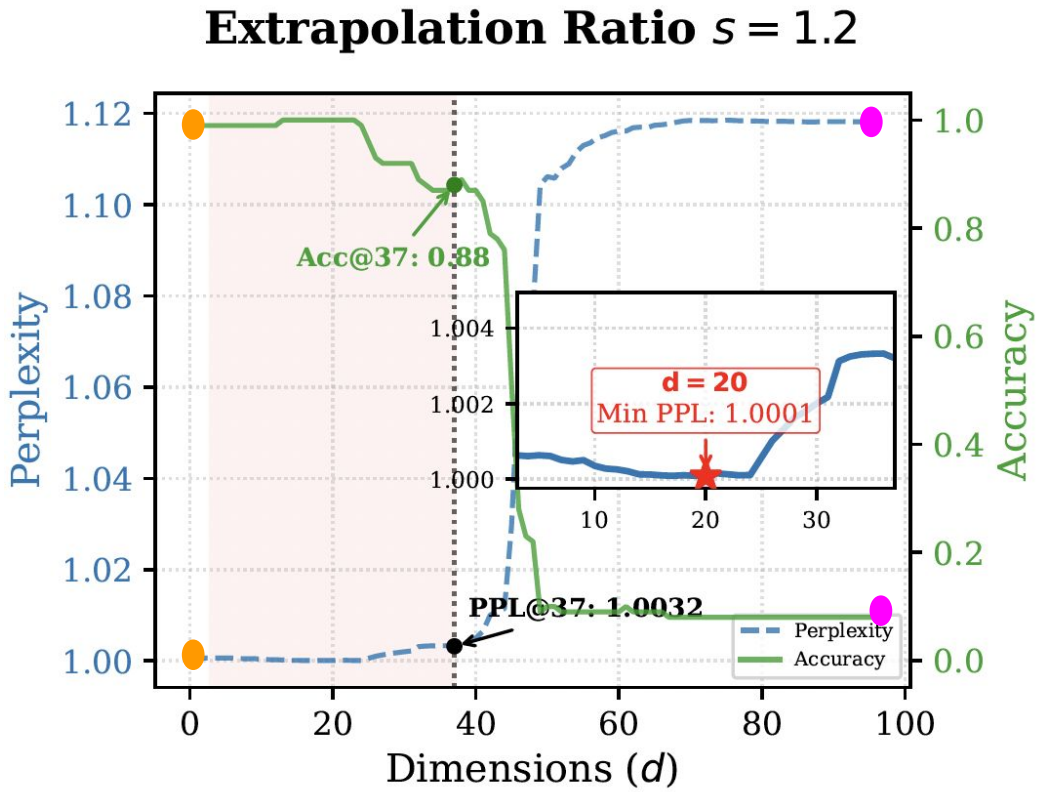
$$\mathfrak{R}_i^*(\mathbf{x}, m) = \begin{cases} \mathfrak{R}_i(\mathbf{x}, m/s) & \text{if } i \in \mathcal{S} \\ \mathfrak{R}_i(\mathbf{x}, m) & \text{if } i \notin \mathcal{S} \end{cases}, \quad s = \frac{L'}{L_{\text{train}}}$$

$\mathcal{S}$: set of indices of RoPE planes that we interpolate.

Analyzing the sensitivity of the model's extrapolation capability to different spectral components of the positions encoding.

Critical Band

$$\mathfrak{R}_i^*(\mathrm{x}, m) = \begin{cases} \mathfrak{R}_i(\mathrm{x}, m/s) & \text{if } i \in \mathcal{S} \\ \mathfrak{R}_i(\mathrm{x}, m) & \text{if } i \notin \mathcal{S} \end{cases}, \quad s = \frac{L'}{L_{\text{train}}}$$



$d = 0$: model performance by interpolating $\mathcal{S}_d = \{0, 1, 2, \cdots, \frac{d_{\max}}{2} - 1\}$

$d = \frac{d_{\max}}{2}$: model performance by interpolating $\mathcal{S}_d = \varnothing$

$d = 37$: model performance by interpolating $\mathcal{S}_d = \{37, 38, 39, \cdots, \frac{d_{\max}}{2} - 1\}$

Exclusive Process:

$$\mathcal{S}_d = \{\, i \mid d \le i \le \frac{d_{\max}}{2} - 1\}, \quad d \in [0, \frac{d_{\max}}{2}]$$

- d_{extra}=37 in this case
- Acc@37 < Acc@20
- Optimal **d**=20 < 37
  - Don't deal with 20 < **d** < 37 cause performance degradation
- **d** with red star: **d_{upper}**
  - **d** < **d_{upper}** is harmful

Critical Band

$$\mathfrak{R}_i^*(\mathrm{x}, m) = \begin{cases} \mathfrak{R}_i(\mathrm{x}, m/s) & \text{if } i \in \mathcal{S} \\ \mathfrak{R}_i(\mathrm{x}, m) & \text{if } i \notin \mathcal{S} \end{cases}, \quad s = \frac{L'}{L_{\text{train}}}$$

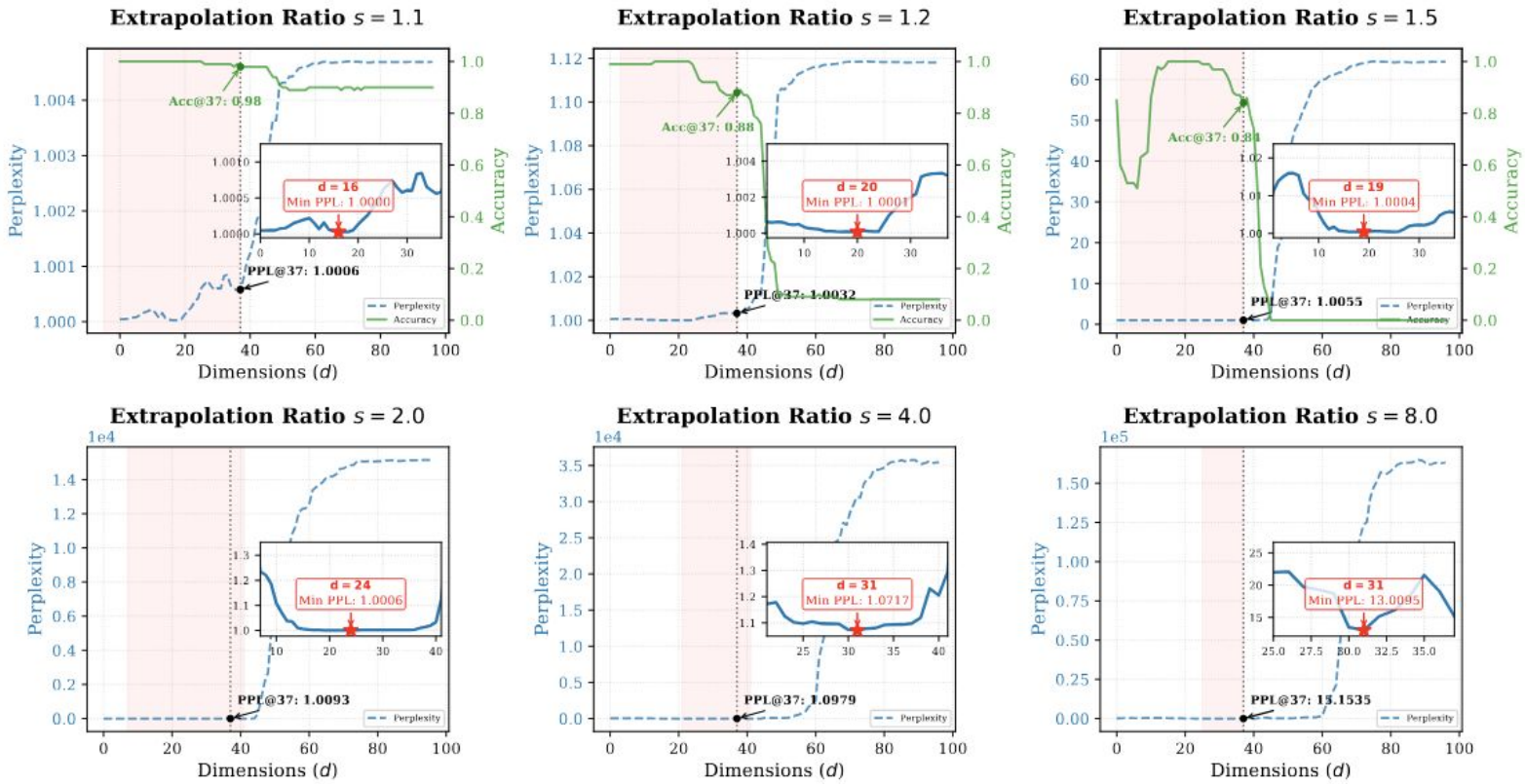


Figure 1: Exclusive-based results of PPL and Accuracy for various extrapolation ratios s for 100-digit string-copying task. Perplexity (PPL) as a function of the interpolation starting index d. The x-axis d represents the upper bound of the dimension range $[d, d_{\text{max}}/2 - 1]$ where interpolation is applied (Equation 6). Shaded regions in the main plots correspond to the magnified insets, providing a detailed view of the local minima of the perplexity for each extrapolation ratio s.

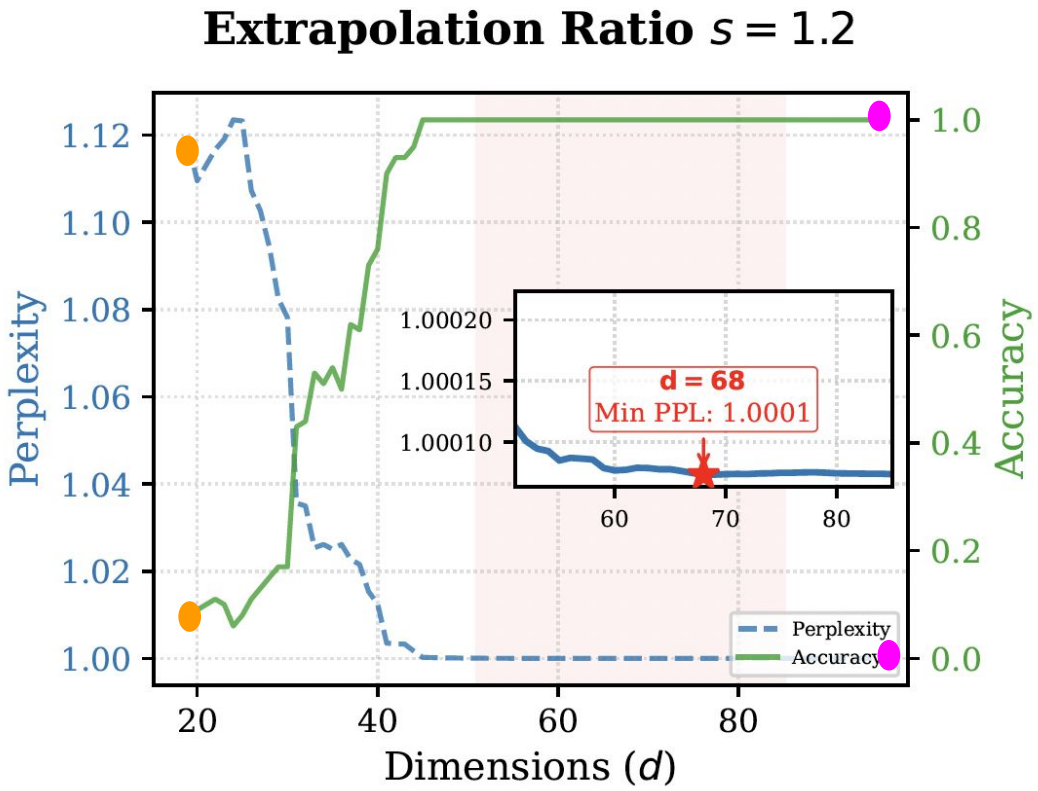
Exclusive Process:

$$\mathcal{S}_d = \{\, i \mid d \le i \le \frac{d_{\text{max}}}{2} - 1\}, \quad d \in [0, \frac{d_{\text{max}}}{2}]$$

As extrapolation ratio $s \uparrow$, $d_{\text{upper}} \uparrow$

Critical Band

$$\mathfrak{R}_i^*(\mathrm{x}, m) = \begin{cases} \mathfrak{R}_i(\mathrm{x}, m/s) & \text{if } i \in \mathcal{S} \\ \mathfrak{R}_i(\mathrm{x}, m) & \text{if } i \notin \mathcal{S} \end{cases}, \quad s = \frac{L'}{L_{\text{train}}}$$



Inclusive Process:

$$\mathcal{S}_d = \{\, i \mid d_{\text{upper}} \le i \le d\}, \quad d \in [d_{\text{upper}} - 1, \frac{d_{\max}}{2} - 1]$$

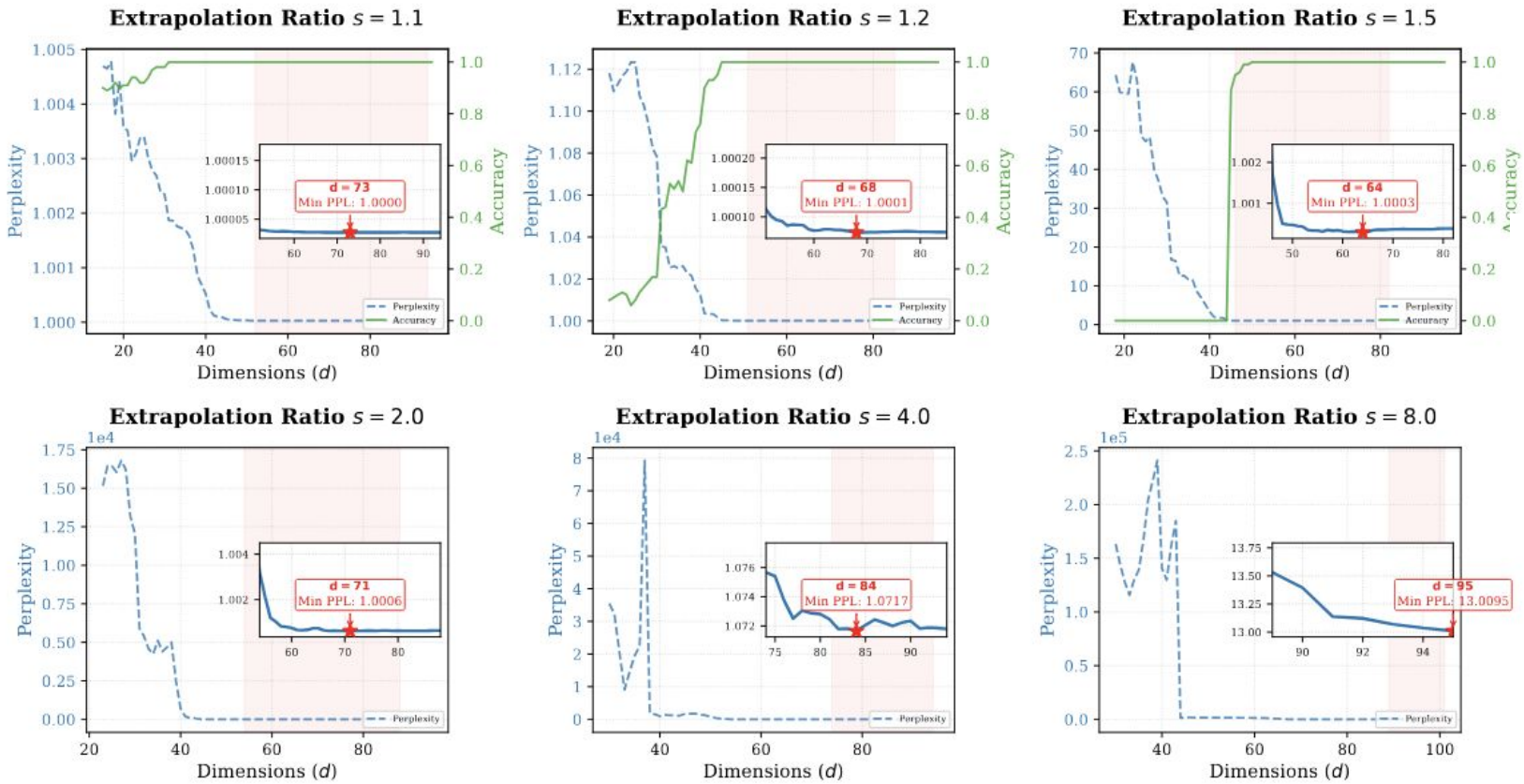
- Optimal **d**=68 < d_{max}/2-1
- **d** with red star: **d_{lower}**
  - ultra-low frequency is **not** harmful

$d = d_{\text{upper}} - 1$: model performance by interpolating $\mathcal{S}_d = \varnothing$

$d = \frac{d_{\max}}{2} - 1$: model performance by interpolating $\mathcal{S}_d = \{d_{\text{upper}}, d_{\text{upper}} + 1, d_{\text{upper}} + 2, \cdots, \frac{d_{\max}}{2} - 1\}$

Critical Band

$$\Re_i^*(\mathrm{x}, m) = \begin{cases} \Re_i(\mathrm{x}, m/s) & \text{if } i \in \mathcal{S} \\ \Re_i(\mathrm{x}, m) & \text{if } i \notin \mathcal{S} \end{cases}, \quad s = \frac{L'}{L_{\text{train}}}$$



Inclusive Process:

$$\mathcal{S}_d = \{\, i \mid d_{\text{upper}} \le i \le d \},\quad d \in [d_{\text{upper}} - 1, \frac{d_{\max}}{2} - 1]$$

As extrapolation ratio $s \uparrow$, $d_{\text{lower}} \uparrow$

Figure 2: Inclusive-based results of PPL and Accuracy for various extrapolation ratios s for 100-digit string-copying task. Perplexity (PPL) as a function of the interpolation ending index d. The x-axis d represents the lower bound of the dimension range $[d_{\text{upper}}, d]$ where interpolation is applied (Equation 7). Shaded regions in the main plots correspond to the magnified insets, providing a detailed view of the local minima of the perplexity for each extrapolation ratio s.

Sliding Critical Band

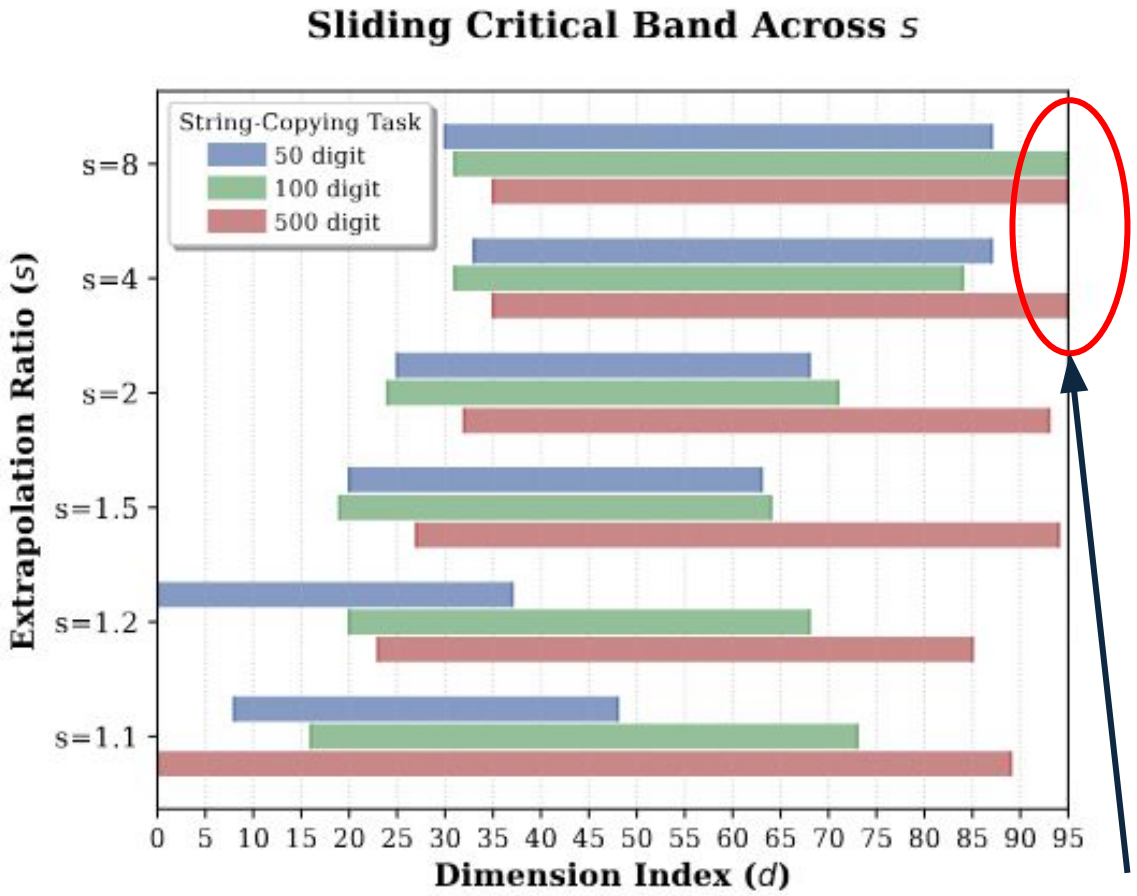


Figure 3: Sliding Critical Band across different task lengths and extrapolation ratios.

$$\text{Critical Band: } \mathcal{S} = \{\, i \mid d_{\text{upper}} \leq i \leq d_{\text{lower}} \}$$

As extrapolation ratio $s \uparrow$, Critical Band $\rightarrow$

Run out of room when $s = 4, 8$ (Green, Red)

Spectrum Bandwidth Exhaustion

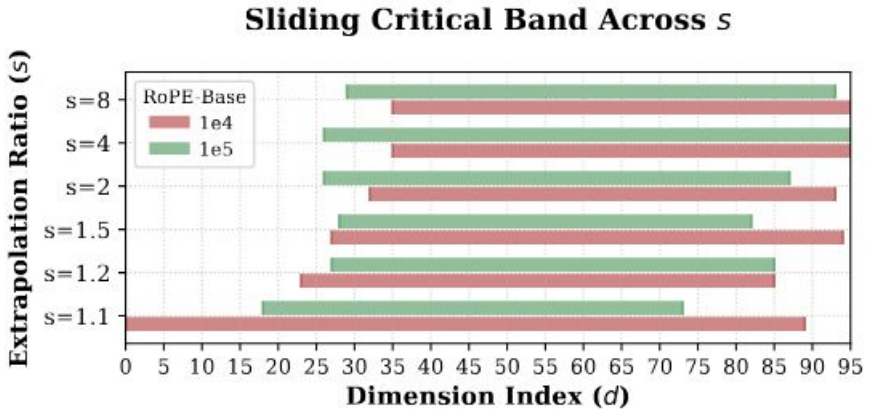


Figure 4: Sliding Critical Band for different RoPE bases with the same task length 500-digit.

Table 3: PPL results after interpolating critical band. Under large-scale extrapolation, larger RoPE-base achieves better performance.

s	$\theta = 10^4$	$\theta = 10^5$
$s = 2$	1.0013	1.0002
$s = 4$	1.0389	1.0178
$s = 8$	3.2016	1.9984

What happens when the extrapolation ratio s increases to a point where the required lower bound d_{lower} exceeds the maximum RoPE-dimension $d_{\max}/2$ of the model?

Performance Collapse

Why does larger θ is effective for context extrapolation?

Llama 3 (Grattafiori et al., 2024), Code Llama (Rozi`ere et al., 2024)

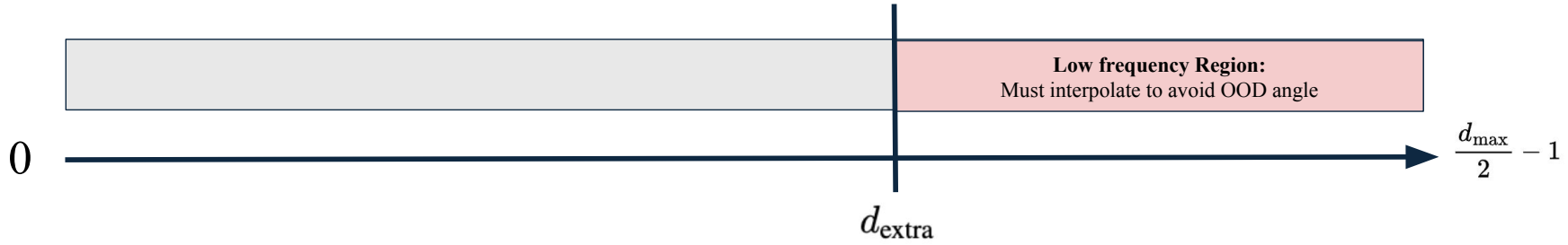
$$\theta \uparrow, \min(\theta_i) \downarrow$$

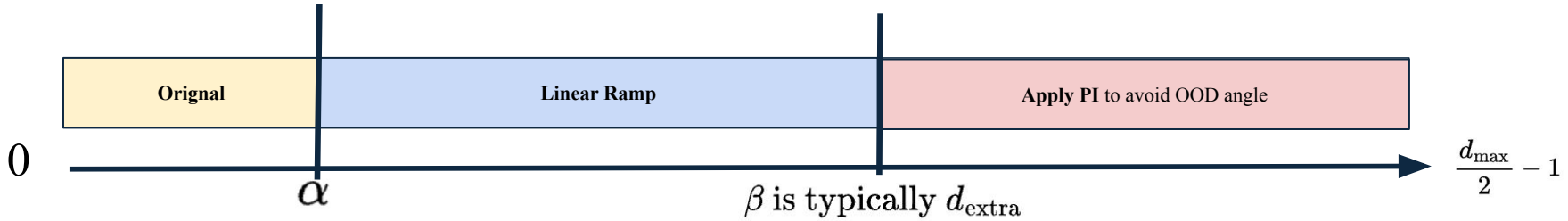
Buying more space for the Critical Band to slide

Same extrapolation ratio s, $\theta \uparrow$,
Critical Band narrower and $\leftarrow$

Sliding Critical Band v.s. Common Belief

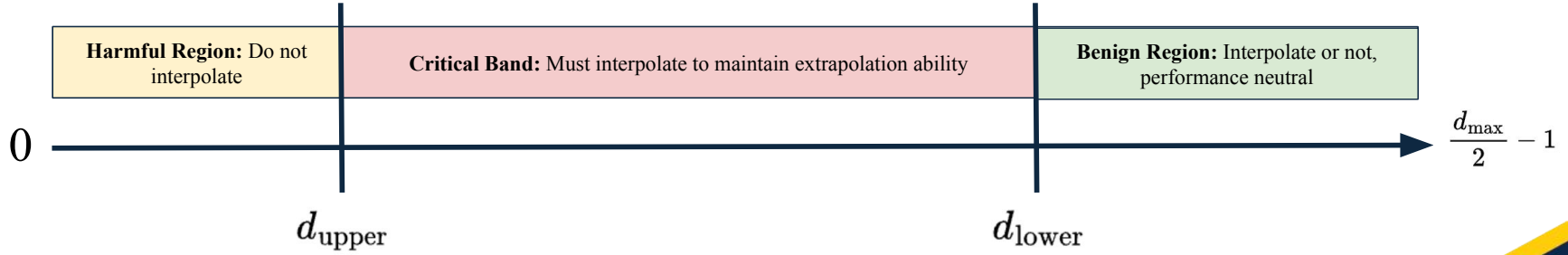
Common Belief:





- ✗ Mild extrapolation ratio
- ✗ Explain larger RoPE-base enhance extrapolation ability

Sliding Critical Band:



- ✓ Mild extrapolation ratio
- ✓ Explain larger RoPE-base enhance extrapolation ability
- ✓ Compatible with existing efficiency methods

Other Perspective: Recovering Slash Pattern Attention

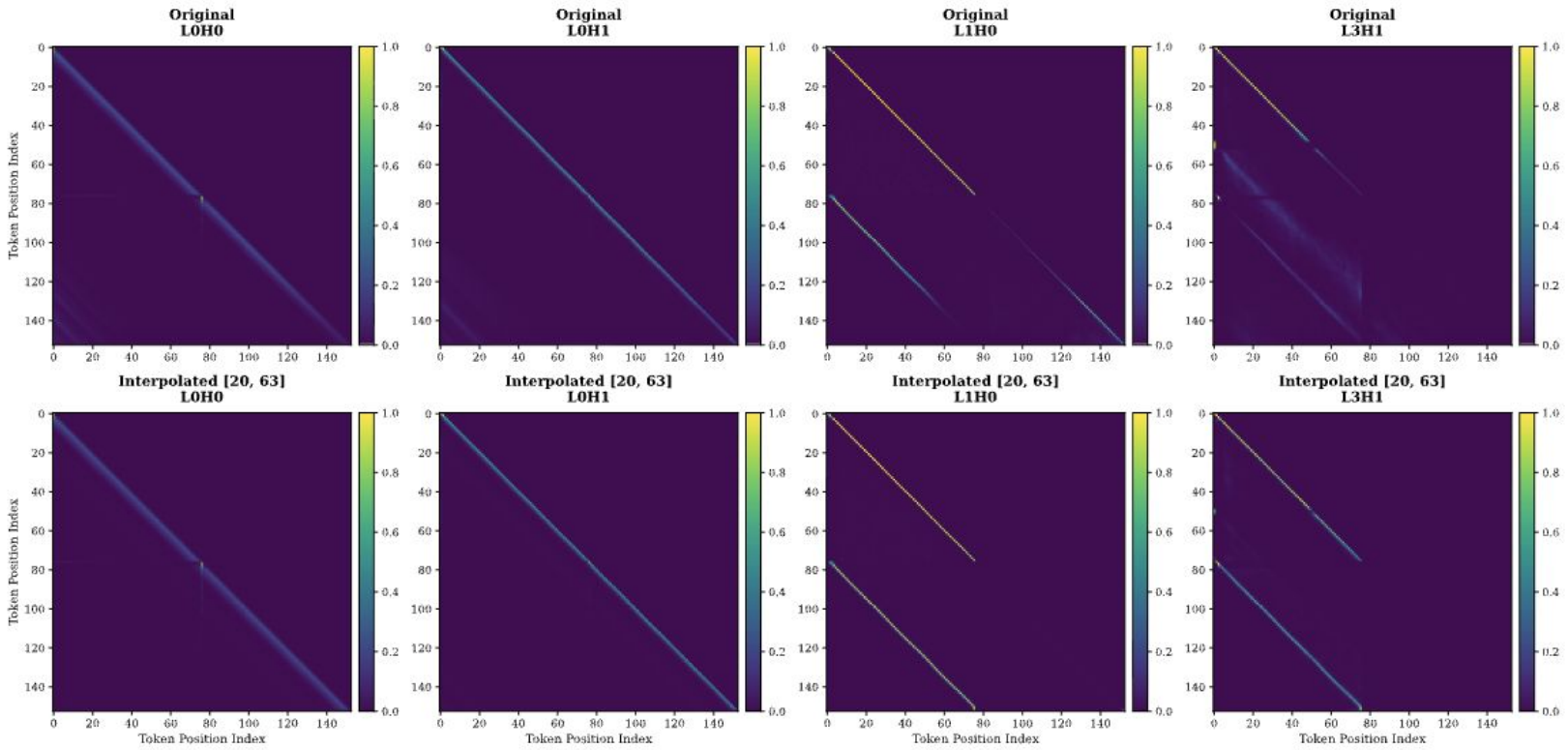
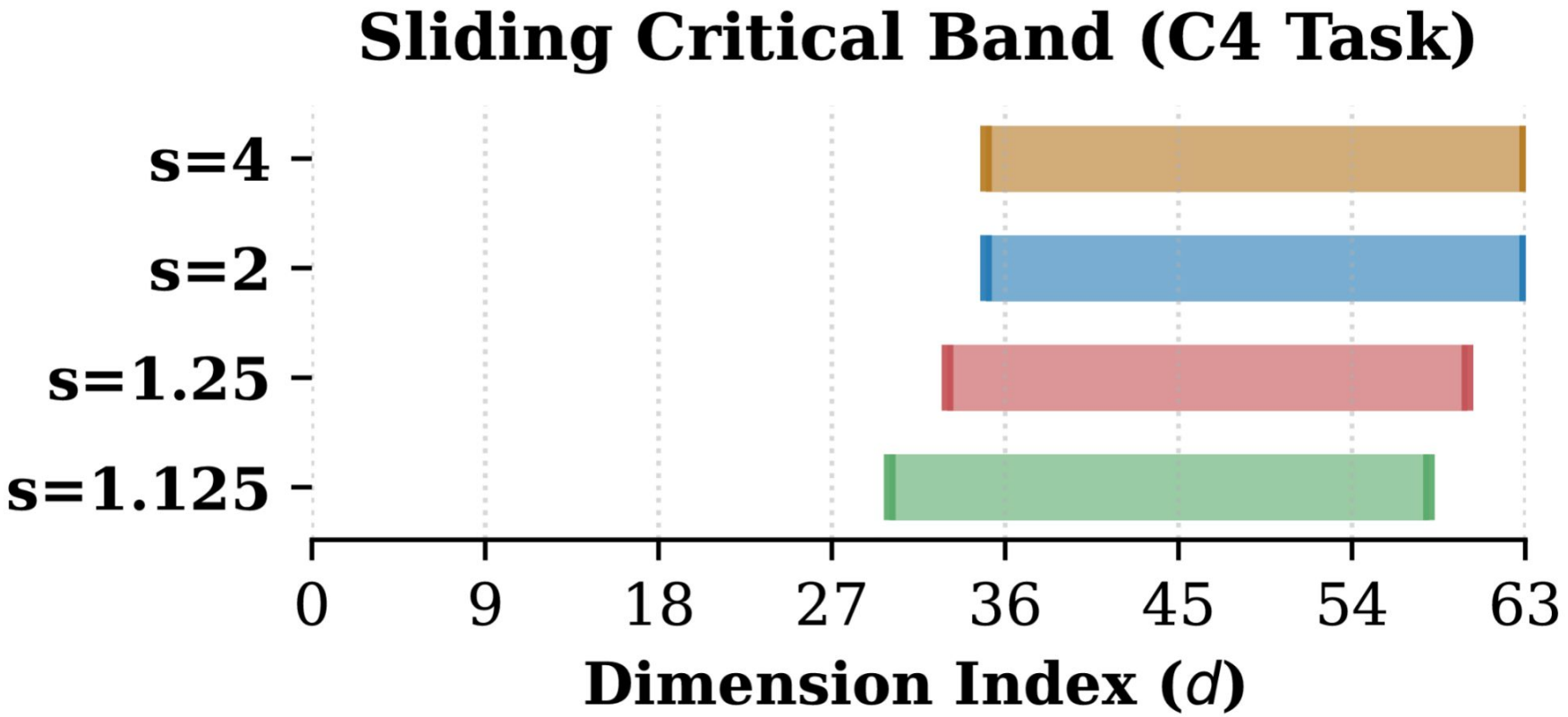


Figure 5: Comparison of attention pattern between original RoPE (first row) and Critical Band interpolation RoPE (second row) in 100-digit string-copying task with extrapolation ratio $s = 1.5$. The a-th head at b-th layer is denoted as LbHa. The slash attention pattern in the first row is weaker than that in the second row.

- Slash-Dominant Heads (SDHs) (Cheng et al., 2026) : attention distribution is primarily determined by RoPE and is largely unaffected by semantic content.

Further Validation: Real-World Scenarios



Summary

- **Re-evaluating Common Belief:** We discover that certain dimensions that complete a full rotation during pretraining harbor hidden threat: without interpolation, they can cause model collapse. Conversely, some dimensions that enter into out-of-distribution angles during inference are actually benign and performance-neutral.
- **Sliding Critical Band:** We formalize a dynamic framework where the dimensions requiring interpolation migrate across the spectrum with the proportion of extrapolation ratio.
- **Spectrum Bandwidth Exhaustion:** Our analysis reveals the mechanism underlying RoPE's base scaling: an increased RoPE base postpones the exhaustion of the available spectrum bandwidth, thereby enhancing the model's capacity for long-context extrapolation.

Future Plan

Diffusion + Language Model: Diffusion LM

- ✓ More intuitive: Human thinking is not that linear like ARM
- ✓ Avoiding error accumulation
- ✓ Some ambitious ideas: Could become a gateway to “real multimodal”

- ? Reducing refine steps
- ? Generalization problems

Miscellaneous

- Accordionist for over 15 years @ Beijing
- Travelled to 20+ countries, 150+ cities





Accordion World Cup @ Shenzhen, China